Shivam Maurya

Embedded Systems Engineer | Firmware, IoT and Edge AI

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SUMMARY

Embedded Systems Engineer (B.E., CGPA 8.06 / 10) with hands-on experience in C/C++ firmware for ESP32, ESP8266, STM32, and Arduino, bus protocols (UART, SPI, I2C, PWM, DMA), and low-power IoT design. Built a 99% deep-sleep-duty-cycle cellular asset tracker, an INT8-quantized 111KB wake-word model running on an ESP32-S3, and a ROS 2 hardware interface with over 1 kHz deterministic UART control loops on STM32. Strong in telemetry, root-cause failure analysis, and real-time control for IoT and robotics hardware. All India Rank under 20, ABU Robocon; 15+ merged open-source pull requests in ros2_control and navigation2.

TECHNICAL SKILLS

Embedded and Firmware: Embedded C/C++ (C++17), C, Firmware Development, Arduino Framework, Interrupt Handling, Hardware Timers, DMA, EEPROM Persistence, Low-Power Design (Deep Sleep), Real-Time Control, PID Control, Watchdog and Fail-Safe Design, Telemetry and Logging

Microcontrollers and Hardware: ESP32, ESP32-S3, ESP8266, STM32 (F4, H7), Arduino, NVIDIA Jetson, SIM800L GSM Module, Digital Electronics, Microcontrollers and Microprocessors

Protocols and Interfaces: UART, SPI, I2C, I2S, PWM, ADC, Custom Binary Protocols (CRC8 framing), MQTT, WebSockets, Eclipse Zenoh-Pico, Micro-ROS, DDS (FastDDS), SMS / GSM, REST APIs

Edge AI: TensorFlow Lite Micro, INT8 Quantization, MFCC Feature Extraction, ONNX Runtime, TensorRT, OpenCV, CPU-Only Inference Optimization

Robotics and Middleware: ROS 2 (Humble, Jazzy), ros2_control, hardware_interface, Micro-ROS, Nav2, Gazebo, NVIDIA Isaac Sim

Tools and Languages: Python, Bash, CMake, Git, GitHub, Linux, Docker, FastAPI, pytest, CI/CD, Root-Cause Failure Analysis

Foundations: Data Structures and Algorithms, Object-Oriented Programming, Probability and Statistics, Database Management Systems

PROJECTS

NanoTracker-24/7: Ultra-Low-Power Asset Tracker | ESP8266, SIM800L GSM, C/C++, EEPROM | github.com/ShivamMaurya14/NanoTracker

- Engineered ESP8266 firmware with a 99% deep-sleep duty cycle (sub-mA sleep current) that locates assets through cellular tower triangulation (no GPS) using a SIM800L module, supporting indoor and underground tracking.
- Implemented SMS-based over-the-air configuration with EEPROM persistence (MODE:TRACK at 60 s, MODE:SAVE at 3600 s intervals) so reporting changes need no physical device access.
- Added dual reporting to the asset owner and logistics company, plus battery voltage telemetry.

Low-Latency Edge Voice Activator for Home Automation | ESP32-S3, TFLite Micro, MFCC, Zenoh-Pico, WebSockets | Smart India Hackathon 2026 (ISRO PS-26172)

- Deployed an INT8-quantized binary wake-word CNN (MFCC to TFLite, 111KB) on an ESP32-S3 using under 10.1% of Flash and 8.2% of RAM, with cloud Vosk speech recognition and a 3-stage NLP intent parser reaching over 97% accuracy on noisy input.
- Implemented dual transport protocols: WebSockets (about 14 ms on LAN) and Eclipse Zenoh-Pico (about 1-2 ms on LAN, 5-byte wire overhead, about 0.9% ESP32 RAM), and drove a 5-channel relay output.
- Built a desktop simulation suite (GUI ESP32 hardware simulator and server manager) for full system testing without physical hardware.

ROS 2 Dual-Mode Motor Interface for STM32 and Arduino | ros2_control, STM32 (F4/H7), Arduino, Micro-ROS, C++ | github.com/ShivamMaurya14/ros2_control_motor_interface

- Developed a ros2_control SystemInterface plugin with dual-mode communication: a custom CRC8-checked binary UART protocol (over 1 kHz deterministic loops) and Micro-ROS native DDS, exposing position, velocity, and effort interfaces.
- Designed hardware-abstracted firmware templates for STM32 (F4/H7) and Arduino with DMA-enabled UART, 1 kHz hardware timer interrupts, encoder mode, and real-time PID loops.
- Added a mock-hardware toggle so the same launch configuration runs on physical hardware or in simulation with no code changes.

Industrial IoT Predictive Maintenance Platform | ESP32, FastAPI, Random Forest, Streamlit | github.com/ShivamMaurya14/Predictive-Maintenance-System | Live demo

- Built live ESP32 sensor ingestion (air and process temperature, RPM, torque, tool wear) through a decoupled FastAPI microservice feeding a Streamlit dashboard with health indicators and failure alerts.
- Trained a Random Forest on the AI4I 2020 dataset reaching over 98% failure-prediction accuracy, with real-time feature engineering and 4 operating modes including a 9-machine fleet simulation.

PCB Fault Detector: Edge Optical Inspection | YOLO, TensorRT, OpenCV, FastAPI, NVIDIA Jetson | github.com/ShivamMaurya14/PCB-Fault-Detector

- Built a plug-and-play PCB defect classifier running YOLO models as .pt or TensorRT .engine files for edge inference, with browser webcam streaming, QR code traceability, and automated CSV inspection logging.

EXPERIENCE

Physical AI Intern | Nextup Robotics | Remote | Aug 2026 - Oct 2026

- Tuned FastDDS (UDP) middleware for zero-packet-drop synchronization between MoveIt 2 and Isaac Sim across a 3-arm robot workcell.
- Integrated tactile contact sensing (7 rigid bodies per robot) and 20 calibrated cameras, scaling the simulation to 30 UR10e robots at 40-50 FPS on a single consumer GPU, and authored 7 engineering guides covering setup, calibration, and CI.

Software Lead | Centre of Excellence for AI & Robotics, Army Institute of Technology, Pune | Jun 2024 - Present

- Led software for the ABU Robocon national robot (All India Rank under 20), running real-time CNN perception at 30+ FPS on CPU-only, resource-constrained hardware through ONNX export and a hot-swappable inference backend.
- Directed a 100+ member research cell and organized hackathons, workshops, and project sprints across 5+ teams.

Open-Source Contributor | ROS 2 Ecosystem (ros2_control, ros2_controllers, navigation2) | Jun 2025 - Present

- Authored 15+ merged pull requests; diagnosed a controller manager regression from backtraces, fixed cross-repository CI build failures, and added static type checking to improve code reliability.

EDUCATION

Bachelor of Engineering (B.E.), Automation and Robotics Engineering | Army Institute of Technology, Pune, India | Jun 2024 - May 2028 (Expected) | **CGPA: 8.06 / 10**

Relevant Coursework: Machine Learning, Deep Learning, Probability and Statistics, Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming

HONORS & ACHIEVEMENTS

- All India Rank under 20, ABU Robocon 2025/26 (Lead Software Architect)
- Smart India Hackathon 2026 participant (ISRO PS-26172, Edge Voice Activator)
- Top 500 All India, Amazon ML Challenge 2026 | Runner-Up, Hackathon 2026 | Top Submission, CODECURE 2025, IIT BHU
- State Level Round 2, Avishkar Maharashtra Inter-University Research Convention 2026 | Finalist, IIIT Pune Thinking Machines 2025

Languages: English (Professional Working Proficiency), Hindi (Native)